

The ratios for 30°, 45° and 60°

Three angles whose exact values come from half a square and half an equilateral triangle — and must be known by heart.

LESSON 26

1 × 40 MIN

GEOMETRY 8, TEMA 23

STAGE 9 · BEYOND

LESSON CLOCK

5'

12'

8'

13'

2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Derive the exact ratios for 45° from half a square.
- Derive the exact ratios for 30° and 60° from half an equilateral triangle.
- Recall the table of exact values.
- Use the exact values in calculations without a calculator.

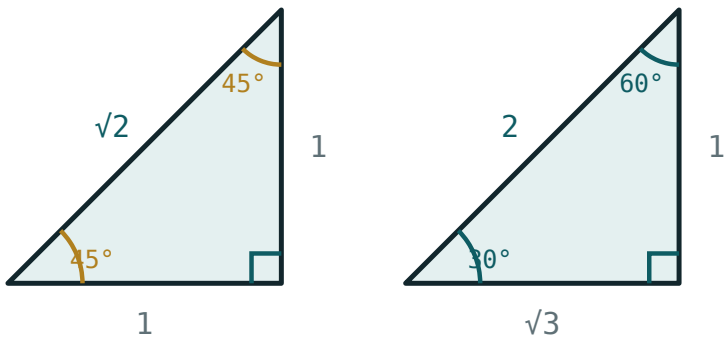
TEXTBOOK

National	Tema 23, pp. 54–55
Cambridge	Extension beyond Stage 9
Workbook	—

01

Explanation

The two triangles that give everything



Half a unit square gives 45°; half an equilateral triangle of side 2 gives 30° and 60°.

For 45°. Cut a square of side 1 along a diagonal. The two legs are 1 and 1, and the hypotenuse is $\sqrt{1+1} = \sqrt{2}$. So

$$\sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \quad \cdot \quad \tan 45^\circ = \cot 45^\circ = 1$$

For 30° and 60°. Cut an equilateral triangle of side 2 down its height. The base is halved to 1, the hypotenuse stays 2, and the height is $\sqrt{4-1} = \sqrt{3}$. The angles are 30° and 60°.

$$\sin 30^\circ = \frac{1}{2} \quad \cdot \quad \cos 30^\circ = \frac{\sqrt{3}}{2} \quad \cdot \quad \sin 60^\circ = \frac{\sqrt{3}}{2} \quad \cdot \quad \cos 60^\circ = \frac{1}{2}$$

The table

A	SIN A	COS A	TAN A	COT A
30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$	$\sqrt{3}$
45°	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1	1
60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$	$\frac{\sqrt{3}}{3}$

HOW TO REMEMBER IT

Write 1, 2, 3 under 30°, 45°, 60°, take the square root of each and divide by 2: $\frac{\sqrt{1}}{2} = \frac{1}{2}$, $\frac{\sqrt{2}}{2}$, $\frac{\sqrt{3}}{2}$.

That is the sine row. The cosine row is the same three numbers backwards.

Approximate values, for checking: $\sin 30^\circ = 0.5$, $\sin 45^\circ \approx 0.71$, $\sin 60^\circ \approx 0.87$.

Where they appear

These three angles run through the whole of school geometry: the height of an equilateral triangle ($h = a \cdot \sin 60^\circ = \frac{a\sqrt{3}}{2}$), the diagonal of a square ($d = a\sqrt{2}$), the

30° slope of a ramp, the shorter diagonal of a rhombus with a 60° angle.

02 Worked examples

EXAMPLE 1 Find $\sin 30^\circ + \cos 60^\circ$

$$\textcircled{1} \quad \sin 30^\circ = \frac{1}{2}$$

From the table.

$$\textcircled{2} \quad \cos 60^\circ = \frac{1}{2}$$

They are equal — 30° and 60° are complementary.

$$\textcircled{3} \quad \frac{1}{2} + \frac{1}{2} = 1$$

ANSWER

1

EXAMPLE 2 A right triangle has hypotenuse 10 and one angle 30° . Find both legs.

$$\textcircled{1} \quad a = c \cdot \sin 30^\circ = 10 \cdot \frac{1}{2} = 5$$

The side opposite the 30° .

$$\textcircled{2} \quad b = c \cdot \cos 30^\circ = 10 \cdot \frac{\sqrt{3}}{2} = 5\sqrt{3}$$

$$\textcircled{3} \quad 5\sqrt{3} \approx 8.66$$

Check: the shorter leg faces the smaller angle ✓

ANSWER

5 and $5\sqrt{3} \approx 8.66$

EXAMPLE 3 Simplify $2 \sin 30^\circ \cdot \cos 60^\circ + \tan^2 45^\circ$

$$\textcircled{1} \quad \sin 30^\circ = \cos 60^\circ = \frac{1}{2}$$

$$\textcircled{2} \quad 2 \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$$

$$\textcircled{3} \quad \tan 45^\circ = 1, \text{ so } \tan^2 45^\circ = 1$$

$$\textcircled{4} \quad \frac{1}{2} + 1 = 1.5$$

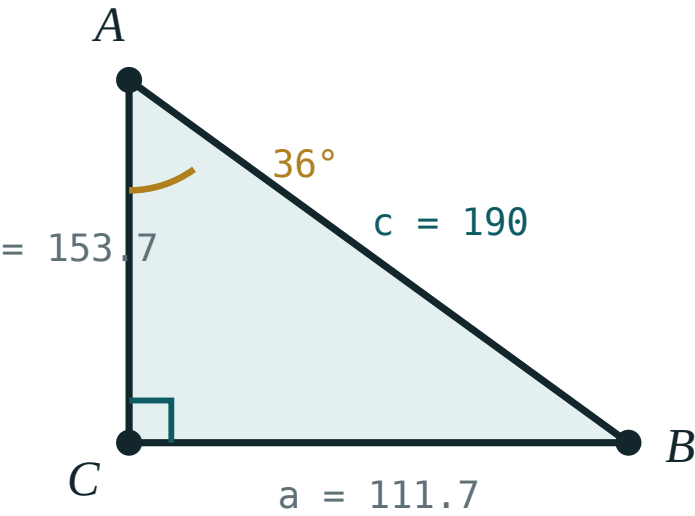
ANSWER

1.5

03 Interactive model

Set the angle to 30° , 45° and 60° in turn and compare the readouts with the table.

Check the table against the model



α 36°	β 54°	a (opposite) 111.7	b (adjacent) 153.7	c (hyp) 190	$\sin \alpha$ 0.59
$\cos \alpha$ 0.81	$\tan \alpha$ 0.73	$a^2 + b^2$ 36100	c^2 36100		

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Exact value	Aniq qiymat	Точное значение
Half of a square	Kvadratning yarmi	Половина квадрата
Equilateral triangle	Teng tomonli uchburchak	Равносторонний треугольник
Table of values	Qiymatlar jadvali	Таблица значений
Special angle	Xos burchak	Специальный угол
Surd	Irratsional ildiz	Иррациональный корень
Derive	Keltirib chiqarish	Вывести
By heart	Yoddan	Наизусть

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$\sin 30^\circ$ equals:

$$\frac{1}{2}$$

$$\frac{\sqrt{3}}{2}$$

$$\frac{\sqrt{2}}{2}$$

$$1$$

$\cos 45^\circ$ equals:

$$\frac{1}{2}$$

$$\frac{\sqrt{2}}{2}$$

$$1$$

$$\sqrt{2}$$

tan 60° equals:

sin 60° equals:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\sin 30^\circ = ?$

ANSWER $\frac{1}{2}$

2. $\cos 60^\circ = ?$

ANSWER $\frac{1}{2}$

3. $\tan 45^\circ = ?$

ANSWER 1

4. $\sin 45^\circ = ?$

ANSWER $\frac{\sqrt{2}}{2}$

5. $\cos 30^\circ = ?$

ANSWER $\frac{\sqrt{3}}{2}$

6. $\tan 60^\circ = ?$

ANSWER $\sqrt{3}$

7. $\cot 45^\circ = ?$

ANSWER 1

Medium · the standard question

7 PROBLEMS

1. $\sin 30^\circ + \cos 60^\circ$

ANSWER 1

2. $\sin^2 45^\circ + \cos^2 45^\circ$

ANSWER 1

3. $\tan 30^\circ \cdot \tan 60^\circ$

ANSWER 1

4. A right triangle has hypotenuse 10 and an angle of 30° . Find both legs.

ANSWER 5 and $5\sqrt{3}$

5. A right triangle has hypotenuse 8 and an angle of 45° . Find both legs.

ANSWER $4\sqrt{2}$ each

6. $2 \sin 60^\circ = ?$

ANSWER $\sqrt{3}$

7. An equilateral triangle has side 6. Find its height.

ANSWER $3\sqrt{3}$

Hard · stretch and reason

7 PROBLEMS

1. Simplify $2 \sin 30^\circ \cdot \cos 60^\circ + \tan^2 45^\circ$

ANSWER 1.5

2. Simplify $\sin 60^\circ \cdot \cos 30^\circ + \sin 30^\circ \cdot \cos 60^\circ$

ANSWER $\frac{3}{4} + \frac{1}{4} = 1$

3. A ramp rises at 30° over a horizontal distance of 12 m. Find its height and length.

ANSWER $h = 12 \tan 30^\circ = 4\sqrt{3} \approx 6.93$, $\ell = 8\sqrt{3} \approx 13.86$

4. A rhombus has side 8 and an angle of 60° . Find its shorter diagonal.

ANSWER 8 — the triangle cut off is equilateral.

5. Find the exact area of an equilateral triangle of side a .

ANSWER $S = \frac{a^2\sqrt{3}}{4}$

6. Simplify $\frac{\sin 60^\circ}{\cos 60^\circ} - \tan 60^\circ$

ANSWER 0

7. A square has diagonal 10. Find its side exactly.

ANSWER $5\sqrt{2}$

07 Homework

Homework — 6 problems

Geometry 8, Tema 23, pp. 54–55. Learn the table before the next lesson.

- Write out the table of values for 30° , 45° and 60° from memory.
- Simplify $\sin 45^\circ \cdot \cos 45^\circ$
- A right triangle has hypotenuse 14 and an angle of 60° . Find both legs.

4. *An equilateral triangle has side 12. Find its height and area exactly.*
5. *Simplify $\tan 45^\circ + 2 \cos 60^\circ$*
6. *A square has side 7. Find its diagonal exactly and to 2 decimal places.*